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FACULTY OF COMPUTER SCIENCE AND ENGINEERING



CO5151 – Advanced Agentic Artificial Intelligence

D2 Interim Report

LuatCheck: A Local, Source-Grounded Assistant over Vietnamese Regulations

Application Track – Group 8

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Repository: https://github.com/tuandung222/LuatCheck-D2

Reported revision: `138128e804bfa99aa03c6fc483920dc96a733225`

Access: public repository (no login needed)

Status: On track; the core system, two 50-question benchmarks, 60-item judge audit and evaluation pipeline are complete, with harder unseen multi-hop and red-team evaluations scheduled for Weeks 45–47 (Section 7).

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Summary. LuatCheck answers a question only from a document collection, cites the sections it used, checks each citation against the source text in code, and refuses or asks for clarification when the sources do not answer. It is a single bounded agent whose controller is a 4-billion-parameter model that runs locally on a laptop (a lighter 2B option is also supported). It uses read-only search and read tools, keyword retrieval with a neural reranker, a deterministic rule-based citation verifier (a Python heuristic gate, not a neural model or second agent) with bounded repair, and input guardrails.

Evaluated (D2 baseline scope): the primary purpose of the D2 evaluation is to verify that the end-to-end system and measurement harness built by the team run reliably and are quantitatively measurable across two benchmarks of 50 Vietnamese questions each (HCMUT regulations, labour law), three configurations (no verifier, verifier, verifier and reranker), three random seeds, and two diagnostic baselines (1,090 recorded trials per controller); broader experiments and system refinements are scheduled for D3. Strict success means appropriate behaviour, a correct and complete answer, and every claim supported by the sources. An LLM judges it (lower bound over all trials); on a blind 60-item human audit (30 per benchmark), judge–human agreement on strict success is 96.7% ($\kappa = 0.93$) on HCMUT regulations and 90.0% ($\kappa = 0.80$) on labour law.

Result: with the 4B controller, the full system reaches 60.7% (HCMUT regulations) and 56.7% (labour law) strict success, compared with 28.7% and 28.0% with the lighter 2B controller (+32.0 and +28.7 percentage points). Without the rule-based citation verifier, 25.6% (HCMUT) and 28.0% (labour law) of answers emitted by the 4B controller leak unverifiable quotes or wrong-clause numbers that slip past the LLM judge; enabling the verifier eliminates unverified citation leakage to 0.0% and raises citation-audited strict success from 55.3% to 60.7–64.0% on HCMUT and from 51.3% to 56.7% on labour law.

1 Changes from the Proposal

Scope and users. The proposal (Application Track) targeted employees and HR staff asking labour-law questions. For this report we added a second corpus, the academic-affairs regulations of HCMUT used by students and advisers, to test whether the same code serves another domain. Wording, tool names and document-id rules are configuration per domain.

We claim only what was evaluated: ten HCMUT documents and ten labour-law documents (the 2019 Labour Code with two laws, six decrees and one circular of 2020–2025), snapshot of 22 October 2026, later amendments unchecked. LuatCheck is not legal advice and has not been tried by real users.

Intended scenarios: a student looks up a rule and opens the cited section; an HR officer checks a labour-law answer against the cited article; a careless or hostile user pastes a document with injected instructions, and the system must keep its tool boundary and refuse or clarify.

Model, retrieval and evaluation adjustments. Instead of the Qwen3 4B/8B controllers and 32B comparator in the proposal, LuatCheck uses two ternary-quantised local models (Bonsai 4B as default and Bonsai 2B as a lighter option) served by `llama.cpp` so that both the live demo and the evaluated controller run on a single 16 GiB laptop. For retrieval, first-stage BM25 and word 1–3-gram TF-IDF fused by reciprocal rank fusion (RRF) feed a Vietnamese cross-encoder reranker (`top_k=5`), with query anchoring on the user's original question on the first search; a dense vector index is implemented and scheduled for ablation in Week 46. Because the proposal's external held-out pair mapping could not be established, we constructed two hash-locked 50-question benchmarks and verified our offline LLM judge with a 60-item blind human audit.

Commitments and their state. Table 1 compares the proposal's commitments due at this checkpoint with their state. We received no feedback on the proposal pitch (22 October), so no change is attributed to feedback.

2 Implemented Architecture

LuatCheck is one agent running a ReAct-style loop [8] over retrieval-augmented generation [6] (Figure 1). Before any model call, input guardrails (regular-expression patterns, shared and per domain; no learned classifier) refuse obviously unsafe input, and the question is wrapped as untrusted text. At each step the controller chooses one action: search, read, answer, clarify or refuse. A search ranks records with BM25 and TF-IDF fused by reciprocal rank fusion (RRF), rescores the top 20 with a neural cross-encoder reranker and returns the best five.

Proposal commitment	State	Change and reason
Budgeted loop, schema output, citation check	Done	Runs end to end; extended with bounded repair and better quote matching.
Qwen3 4B/8B controllers, 32B comparator	Dropped	Replaced by two small ternary models (4B and 2B, Section 2): one 16 GiB laptop runs the demo and the benchmark model.
BM25 + Qdrant dense retrieval; 200 held-out question pairs	Changed	BM25 + TF-IDF with rank fusion and a neural reranker. Dense index implemented, not evaluated. The pair mapping could not be established; two own benchmarks replace it.
≥10 end-to-end tasks, 3 configurations × 3 seeds	Done	100 questions, 3 configurations × 3 seeds, 1,090 trials per system.
Reranker (optional)	Done	In the current system; ablated in Section 5.
Semantic (NLI) check, optional	In progress	Implemented, not used in any measurement yet.
Guarded summary sending, UI, audit log	Done	Changed: the model can only propose; a trusted UI approves; web UI with cited sources and agent timeline.
Verifier audit, 30 pairs	Done	Changed to a blind human audit of the LLM judge (30 items per benchmark, 60 total): 96.7% agreement ($\kappa = 0.93$) on HCMUT and 90.0% ($\kappa = 0.80$) on labour law.
Threat model	Done	Updated in Section 6.
First injection results	Not started	Not required at this checkpoint; planned for week 47.
HR walkthrough	Not started	Scheduled for week 45 (2–8 November; owners: Quyền, Phong).

Table 1: Proposal commitments due at this checkpoint and their state.

Rule-based citation verifier (L1 lexical heuristic, not a model or agent). Every proposed answer goes through a deterministic Python heuristic module (`src/verifier/lexical_grounding_v2.py`, < 2 ms on CPU, zero LLM or neural calls) that enforces three hard gates and one diagnostic check: (i) *retrieval provenance*: every cited `article_id` must exist in the corpus and have been returned by `search` or `read_document` during the current trial; (ii) *normalised verbatim/elided quote match*: after Unicode NFC, case and whitespace/line-wrap folding, every non-trivial segment (≥ 12 characters or ≥ 3 words) of an elided quote (`...`) must appear verbatim and in order inside the cited record, with digit-boundary guards so `5` cannot match inside `15` or `3,5`; (iii) *claim-local number and unit match*: every numeric quantity and unit in an answer sentence must occur inside the ≤ 300-character source window that best matches the surrounding ± 8 context words of that claim; and (iv) *content-word overlap*: if fewer than 50% of a sentence's non-stopword tokens appear in the best source window, a soft diagnostic warning is logged. If any of the three hard gates fails, a structured tool report naming the failed rule and quoting the closest 170-character source span is returned to the controller for at most 2 repairs; if repairs are exhausted, the loop terminates as a refusal. This is deterministic external feedback rather than LLM self-critique [5]. The loop is bounded to 8 steps, 5 tool calls, 2 repairs and 180 s. A backend or format failure ends the run as an execution error; it is never turned into an answer or a refusal.

There is one controller and no multi-agent coordination, which keeps permissions and budgets easy to inspect; the loop, tool registry, verifier and harness are custom Python (no external agent framework). LuatCheck uses the 4B controller by default; a lighter configuration uses the 2B controller. Both are ternary-quantised Bonsai models[1] served by the same server. Everything in Figure 1 except the dashed boxes runs end to end. The two orange components, the reranker and the verifier, have on/off switches and are ablated in Section 5. The dashed boxes are implemented but not used in any measurement. There is no code-execution tool and no external sender.

A real trace and the interface. Figure 2(a) shows one recorded trial of LuatCheck on a labour-law question (3 controller calls, 2 tool calls, no repair). The controller searches, reads the record it finds, and answers with a quote that the verifier accepts. Figure 2(b) shows the web interface answering the

[1]Bonsai 4B and 2B, ternary weights (about 2.1 bits per weight) in GGUF format, served locally by llama.cpp.

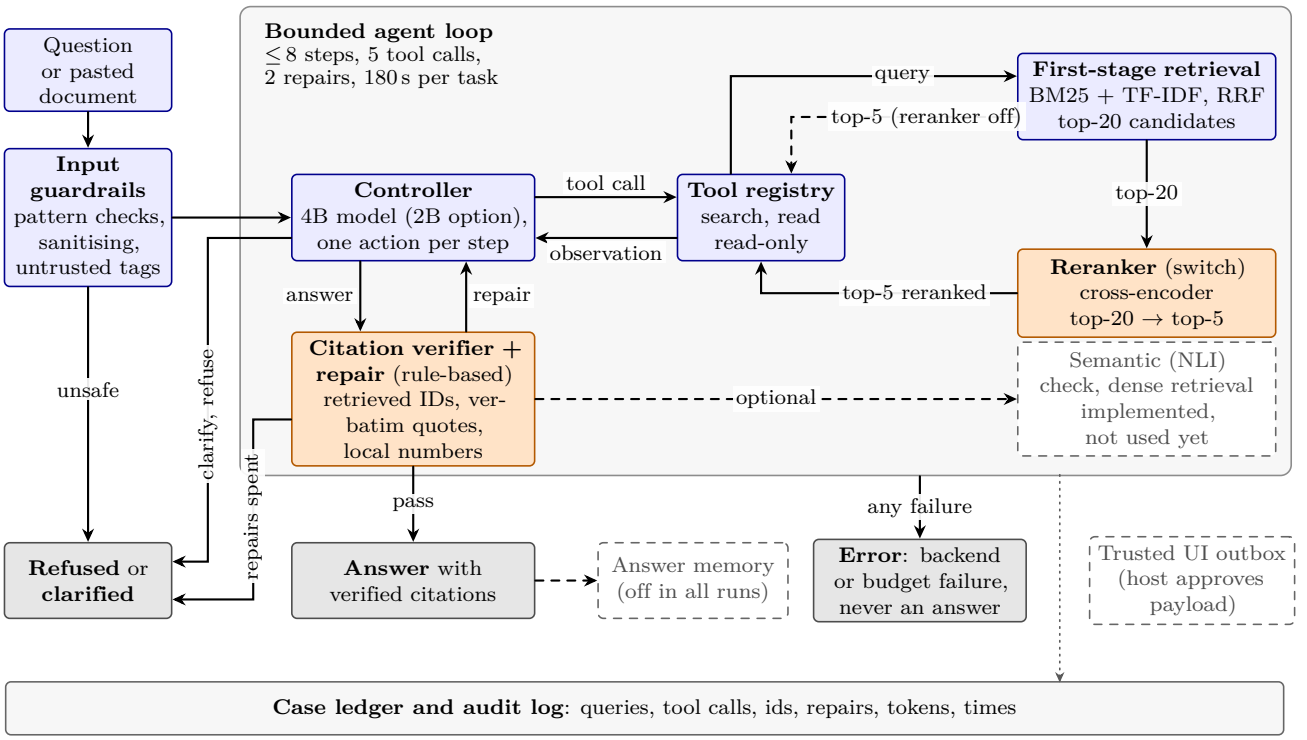


Figure 1: Architecture and control flow of LuatCheck. Blue: always on. Orange: on/off switch, ablated in Section 5. Dashed: implemented, not used in any measurement. Grey: terminal states of a run. The dotted arrow marks that every step is written to the case ledger.

Question (translated): under Decree 293/2025, what is the hourly minimum wage in region IV?

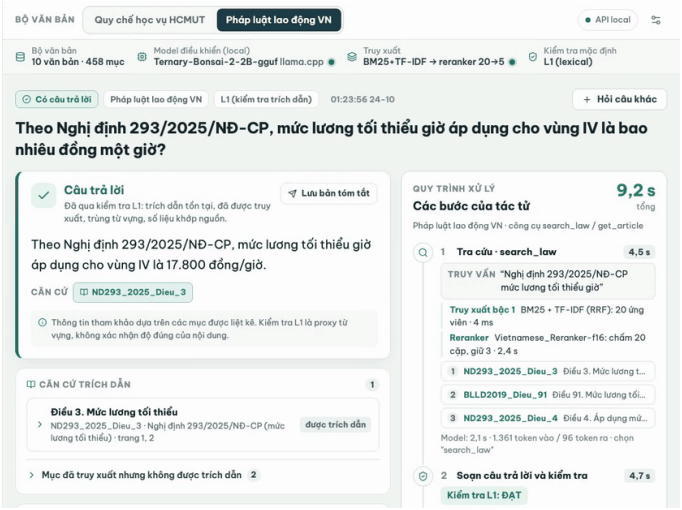
1 search. Query "Nghị định 293/2025/NĐ-CP mức lương tối thiểu giờ vùng IV"; BM25 + TF-IDF give 20 candidates, the reranker keeps 5; Article 3 of the decree is ranked first. *1,574 tokens in, 105 out.*

2 read. Full text of Article 3 of Decree 293/2025. *2,921 in, 166 out.*

3 answer. "17,800 VND per hour", quote "Vùng IV 3.700.000 17.800" from Article 3. **Citation verifier: accepted, 0 repairs.** *3,623 in, 211 out.*

Outcome: answered with a verified citation; 8,118 tokens in, 482 out in total. The answer equals the benchmark's reference answer.

(a) Trace of LuatCheck (4B controller, verifier and reranker on).



(b) Web interface, same question.

Figure 2: (a) One completed trial from the evaluation (a labour-law question about the hourly minimum wage, seed 42). (b) The web interface answering the same question, showing the answer, citation verifier status, highlighted source section and step timeline.

same question, displaying the answer, the verifier verdict, the cited section and the agent timeline. The web interface is the live demo.

3 Tools and Permissions

Table 2 lists every tool and model-facing interface with its state and permission level. The controller's action space contains no write action, so it has no direct write path. Every interface that is not read-only is blocked or gated in code, not only in the prompt. The loop does not use the Model Context Protocol (MCP) server that exposes the same tools; a test of that server confirmed tool discovery, search, lookup, blocked unverified writes and proposal-only summaries. The demo serves one user in one process;

multi-user isolation is not certified.

Interface	State	Permission level	Enforced boundary
Search and read tools (for labour law)	Implemented, used	Read-only	Return local corpus text only.
Reranker (inside search)	Implemented, used	Read-only model component	Scores candidates; a failure is an execution error, never a silent fallback.
Answer-memory write	Implemented, off in all runs	Reversible write	Local database write only after the verification gate.
Case-saving tool of the MCP server	Implemented, blocked	Reversible write	A model caller cannot self-attest verification; the call returns blocked.
Summary-sending tool	Proposal only	Irreversible write	The model-facing tool only creates a proposal; it cannot approve or execute.
Trusted-UI outbox	Implemented	Irreversible write (local queue)	Needs host approval and a token bound to the payload hash; a queue receipt is not a delivery.
External sender, code execution	Not implemented	Irreversible write, code execution	No e-mail or webhook transport and no code-running tool exists.

Table 2: Tool inventory with permission levels as classified in code.

4 Memory and Knowledge Sources

Short-term context. Each run keeps one message list: the domain system prompt, the question (and any pasted document) wrapped as untrusted input, then alternating action and observation turns. Observations are compacted: a search returns the best-matching clauses of each record with a truncation flag, a repeated query or read returns a short notice, and budget notices tell the model when to answer or refuse. The controller has an 8,192-token context window. A case ledger records queries, retrieved ids, repairs, token counts and times.

Long-term memory (implemented, not used in any run of this report). Verified answers can be stored in a local database after the verification gate. Reuse requires a matching index version and a question similarity of at least 0.85, and the reused answer is re-verified. Memory is off in every run of this report and in the demo, so trials stay independent; matching sees only the question and not the facts that decide a case. A fixed set of patterns screens for personal data before a write.

Knowledge sources. Both corpora were built from public PDFs by text extraction, Unicode normalisation and segmentation at section headings. Each record carries its source URL, file hash and page numbers, and the retrieval unit is a record (an article or a clause). The HCMUT corpus has 10 documents and 143 records; five documents are amendment-only and the image-only base regulations are absent, and printed decision numbers come from a web listing. The labour corpus has 10 documents and 458 records; Decree 145/2020 and unnumbered annexes are absent, texts are as promulgated and later amendments are unchecked. Neither corpus has been reviewed by an external legal or academic-affairs specialist.

Retrieval and reranking. The index is built in memory at start-up. BM25 over title and content and TF-IDF over word 1–3-grams are fused by RRF. The top 20 candidates are rescored by a cross-encoder reranker (a Vietnamese reranker model); records longer than one window are split into overlapping windows of about 1,500 characters and the record keeps its best window. The first search of a run always uses the user's own question together with the model's query. Dense embedding retrieval with a stored vector index is implemented but not evaluated; it is a planned ablation for Week 46.

5 Evaluation Set and First Results

Evaluation objective and D2 scope. The primary objective of the evaluation in this D2 report is to demonstrate that the end-to-end agentic pipeline and automated evaluation harness built by the team run reliably and produce reproducible, auditable measurements across configurations. Substantial room for improvement and additional experiments remain—including harder unseen multi-hop queries where keyword retrieval is not saturated, reducing verifier repair-budget over-refusals, evaluating dense retrieval, and running red-team security tests—which will be carried out in **D3** (Section 7).

Benchmarks. Each self-built benchmark has 50 Vietnamese questions over its corpus (100 now; D3 adds 50–100 unseen harder questions and ≥30 attack cases, Section 7): 40 single-record, 5 two-record and 5 out-of-scope questions to refuse. Each comes with gold records, a reference answer and verbatim quotes, locked by hash before collection.

An AI model drafted the questions; two independent AI verifiers and an AI adjudicator cross-checked them; mechanical checks confirmed that every quote is verbatim. Then group member Võ Phạm Tuấn Dũng reviewed all 100 questions on 23 October and requested no change. They are AI-written questions over real documents, not real users' questions. A quality audit found no wrong gold answer and three limits:

- plain BM25 on the raw question already ranks a gold record first for 89% (HCMUT regulations) and 78% (labour law) of questions, so retrieval cannot separate the configurations;
- about one third of the answerable questions can be answered without the corpus (LLM estimate);
- 22 HCMUT questions pin a document by a decision number taken from a web listing.

How success is decided. *Strict success* counts a trial only if the run completed, the behaviour was appropriate (answer when the corpus answers, refuse when it does not), the answer is judged correct and complete, and every factual claim is supported by the sources. Execution errors and failed judgments count as failures, so the rate is a lower bound over all planned trials. It is an LLM estimate, not a human label.

Three measures need no judge: behaviour match, over-refusal and unsafe answers. Recall@5 (are the gold documents among the agent's results?) and citation F1 (are the cited documents the gold ones?) are measured on document ids. They say nothing about answer correctness. Intervals are 95% bootstrap intervals over questions, with the three seeds averaged per question. With 50 questions, differences below about 10 percentage points are not detectable. Definitions are in the repository.

LLM judge and its check. An LLM judge (DeepSeek V4 Flash, through an API gateway) rates each output blind to the configuration and verifier outcome (offline only; model revision not fixed [2]; LLM judges can be biased [9]). On a blind 60-item human audit (30 per benchmark, 4B outputs), strict-success agreement is 96.7% (29/30, $\kappa = 0.93$; weighted 98.8%, $\kappa_w = 0.97$) on HCMUT and 90.0% (27/30, $\kappa = 0.80$; weighted 97.5%, $\kappa_w = 0.95$) on labour law; behaviour agreement is 100.0% and 93.3% ($\kappa = 0.84$). All four disagreements are documented in the repository. Separating correctness from citation support follows Gao et al. [3].

Design. The full system (verifier and reranker) is compared with two simpler configurations: *no verifier* (neither verifier nor reranker) and *verifier* (verifier with repair, no reranker). Each runs three seeds on 50 questions per benchmark.

Two diagnostic baselines (one seed; they are not system variants) bracket the results: *closed-book* (the question only, no tools or documents) and *oracle* (the gold records given up front, answerable questions only). One controller thus has 900 system trials plus 190 baseline trials, 1,090 in all, every one recorded. Both the 4B controller (default) and the 2B controller (lighter option) are evaluated under the same protocol. Accuracy was collected on a shared server GPU (one NVIDIA H100). Latency is measured separately on an Apple M4 laptop (16 GiB), one query at a time.

What the numbers say.

(1) Controller scale drives answer synthesis and citation quality (Figure 3). For the full system (*verifier + reranker*), strict success is 60.7% [50–71] on HCMUT regulations and 56.7% [47–67] on labour

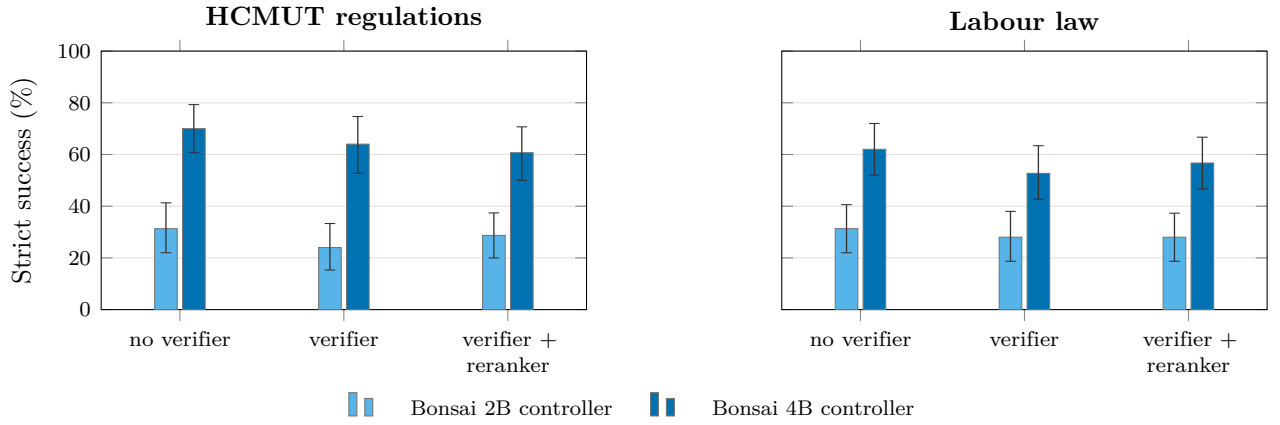


Figure 3: Strict success (LLM-judge lower bound; bars: mean over three seeds; whiskers: 95% bootstrap interval over the 50 questions) across the three configurations for the Bonsai 2B and Bonsai 4B controllers. The right-most group, *verifier + reranker*, is the full system.

law with the 4B controller, compared with 28.7% [20–37] and 28.0% [19–37] with the 2B controller.

Scaling the controller from 2B to 4B on the same pipeline adds +32.0 percentage points [21.3, 42.0] on HCMUT and +28.7 [18.0, 39.3] on labour law for the full system (+38.7 and +30.7 without verifier; +40.0 and +24.7 with verifier only). All six paired bootstrap differences across the three configurations and two benchmarks exclude zero.

(2) Why the citation verifier is essential: eliminating unverified citation leakage and raising citation-audited accuracy. In Table 3, if we look only at the LLM judge's score (*judge-only strict success*, which evaluates answer prose without checking whether the quoted string exists verbatim in the source PDF), *No verifier* scores 70.0% and 62.0% vs 60.7% and 56.7% for the full system because the verifier refuses when two repairs still fail.

Table 4 reveals what happens under the hood: without the verifier, **25.6% (31/121) of HCMUT answers and 28.0% (37/132) of labour-law answers emitted by the 4B controller** (and 39.8% and 45.9% on the 2B controller) **leak unverifiable citations**—altered or fabricated quotes that cannot be highlighted in the PDF, unretrieved section IDs, or numbers borrowed from the wrong clause. Because the LLM judge does not run substring or window-number verification against the full corpus, it awards strict success to 22 (HCMUT) and 16 (labour law) *No verifier* answers that carry broken citations. When we require both LLM-judge correctness and a verifiably authentic citation (*Audited strict* in Table 4), enabling the verifier **eliminates unverified citation leakage to 0.0%** (0/109 and 0/108), rescues 17–20 flawed first drafts via bounded repair, and **raises citation-audited strict success from 55.3% to 64.0% (+8.7 pp, Verifier) and 60.7% (+5.4 pp, Full) on HCMUT, and from 51.3% to 56.7% (+5.4 pp, Full) on labour law**. Recall@5 is already 0.98–0.99 without the reranker on these 100 questions; we retain both components and will test the reranker on a harder unseen benchmark in Week 45.

	HCMUT regulations		Labour law	
Configuration (4B controller)	All 50 questions	45 answerable	All 50 questions	45 answerable
Closed-book (no documents)	2.0 [0–6]	0.0 [0–0]	24.0 [12–36]	24.4 [13–38]
No verifier, no reranker	70.0 [61–79]	69.6 [60–79]	62.0 [52–72]	60.7 [50–71]
Verifier (with repair)	64.0 [53–75]	61.5 [50–73]	52.7 [43–63]	51.1 [40–62]
Verifier + reranker (full system)	**60.7 [50–71]**	59.3 [49–70]	**56.7 [47–67]**	54.1 [43–64]
Oracle (gold records given up front)	–	82.2 [71–93]	–	75.6 [62–87]

Table 3: Strict success (%, LLM-judge lower bound, 95% bootstrap interval over questions) of the D2 system with the 4B controller. Configurations: three seeds; closed-book and oracle: one seed. The oracle runs only on the 45 answerable questions, so the right-hand columns of each benchmark compare every row on the same 45 answerable questions.

(3) Baselines. Without documents the 4B model scores 2.0% [0–6] on HCMUT regulations and 24.0%

Benchmark	Configuration	Unverified cit. leak (%)	Audited strict (%)	Behaviour match (%)	Over-refusal (%)	Faithful-ness (%)	Citation F1	Recall@5
HCMUT regulations	No verifier	25.6	55.3	88.0	9.8	89.8	0.820	0.989
	Verifier	**0.0**	**64.0**	82.7	17.4	90.3	0.704	0.989
	Verifier + reranker	**0.0**	**60.7**	80.0	18.9	89.3	0.714	0.978
Labour law	No verifier	28.0	51.3	94.0	4.4	87.3	0.841	0.989
	Verifier	**0.0**	52.7	81.3	16.8	86.3	0.728	0.989
	Verifier + reranker	**0.0**	**56.7**	79.3	19.1	85.6	0.691	0.993

Table 4: Ablation of the rule-based citation verifier and the reranker (4B controller, 3 seeds × 50 questions per row). *Unverified cit. leak*: share of emitted answers whose quotes, section IDs or numbers fail deterministic source verification against the full PDF corpus (31/121 HCMUT and 37/132 labour answers in *No verifier* vs 0/109 and 0/108 with the verifier). *Audited strict*: trials that pass both the LLM judge and deterministic L1 citation verification. Behaviour match, Over-refusal, Citation F1 and Recall@5 need no judge.

[12–36] on labour law, so part of the labour-law score can come from the model's own knowledge. With the gold records in front of it, the oracle reaches 82.2% [71–93] and 75.6% [62–87] on the answerable questions. Even with perfect evidence the 4B controller misses strict success on about one question in five, against 69.6% and 60.7% for the agent without verifier on the same questions. 17 of 1,090 trials ended as execution errors (wall-time limit); they count as failures.

Error analysis and experimental roadmap for D3. Three failure modes account for the remaining errors in the 4B full system: (i) *repair-budget refusals* (13/150 HCMUT and 18/150 labour trials refused after two L1 quote/number rejections on answerable questions), (ii) *multi-clause synthesis omissions* (oracle gap of about 18–24% even when gold articles are given up front), and (iii) *wall-time budget timeouts* (17 of 1,090 trials). Because D2 evaluation focuses on establishing a working, measurable baseline and lexical retrieval is near ceiling on this initial 100-question set, D3 will target these bottlenecks and evaluate generalization on harder unseen multi-hop and paraphrased benchmarks, repair-loop improvements, and entry-point attack tests.

Cost and latency. The controller and reranker run locally, so there is no per-call API cost. For the full system the controller reads on average 12,070 (HCMUT regulations) and 11,477 (labour law) tokens and writes 763 and 865 tokens per trial, over 3.4 and 3.7 model calls. The judge cost, as reported by the gateway and not an invoice, was USD 1.06 and USD 1.10 for the two 545-trial runs. Collecting a run took 34 and 33 minutes of wall time on the shared H100 (GPU-active hours were not measured). On the laptop (Apple M4, one question at a time, 50 questions per benchmark), the full system answers in a median of 33.0 s (HCMUT regulations) and 29.8 s (labour law) per question with the 4B controller (95th percentile 86 s and 72 s), and in 13.3 s and 14.6 s with the 2B controller (95th percentile 30 s and 28 s); the 4B model generates about 43 tokens/s, the 2B about 92.

Reproducibility. Repository: `https://github.com/tuandung222/LuatCheck-D2`; reported commit `138128e804bf`; full generated per-run result tables are in `deliverables/d2/report/appendix_tables.pdf` and provenance records in `RESULTS_PROVENANCE.md`. All numbers and tables are regenerated from recorded runs by one command (`make report-check`).

6 Threat Model

Assets. (A1) Faithfulness: answers and citations that users rely on must trace to source sections [7]. (A2) User data: questions, pasted documents, memory and the audit log. (A3) Integrity of corpora, domain profiles, prompts, model weights and the frozen benchmark. (A4) The tool boundary: no unreviewed write or egress. (A5) Availability of a bounded local machine.

Attacker. The attacker can submit arbitrary questions and pasted documents through the interface or API (instruction override, delimiter spoofing, prompt or memory exfiltration, requests to evade a regulation), including text copied from a hostile third party. The attacker cannot change code, corpora, configuration, weights, the ledger or the approval state of the trusted UI, and cannot reach loopback servers or other users' sessions. Non-goals: a compromised host or supply chain, poisoned official sources,

multi-tenant isolation, and the correctness of the regulations themselves.

Entry point (asset)	Mitigation	Residual risk
Question, pasted document (A1, A4)	Regex guardrails (shared and per-domain patterns) before any model call, Unicode and delimiter sanitising, untrusted-text tags, fixed action space without a write action	Paraphrases bypass patterns; Vietnamese and English only; false positives; no learned classifier; not attack-tested yet.
Retrieved text (A1)	Observations are tagged as tool results; the citation verifier checks ids, provenance and quotes	Tagged, not filtered: an instruction inside a source document would reach the model.
Ingested documents (A3)	Offline, manual ingestion; source URL, hash and pages per record; benchmark locks hash the corpora	No signature check; trust in the official site.
Memory (A1, A2)	Off in all runs and in the demo; index version, similarity, re-verification, personal-data patterns	Poisoning by an earlier user; incomplete personal-data filter.
Weights, configuration, binaries (A3)	Recorded hashes and commit, identity checks before each run, loopback-only endpoints	Third-party weights and the llama.cpp fork are not audited.
Judge gateway (A2)	Offline only; benchmark data only; key kept in the environment; blind packets; output validation; human audit	Candidate text could steer the judge (untested); model version not fixed.
Web interface, API, outbox, long inputs (A4, A5)	Loopback bind, session cookie and CSRF token, size limits; proposal-only tools with host approval; step, tool, repair, time limits	Single-user assumption; limits turn overload into errors, not into defences.
Web pages	Not applicable: there is no browsing tool.	—

Table 5: Threat model by entry point. Mitigations are implementation boundaries, not measured attack effectiveness. The full-system injection tests [1, 4] belong to the final report (planned for week 47; no attack numbers in this report).

7 Weekly Plan to the First Final Presentation

Table 6 runs to the first final presentation in week 49. The final phase focuses on harder unseen multi-hop questions, real-user feedback from the HR walkthrough, and entry-point prompt-injection testing.

Week (dates)	Owner	Deliverable and acceptance
24–25 Oct (Done)	Dũng, all	Blind human audit of the 60-item judge sample (30 per benchmark); agreement (96.7% and 90.0%) and κ (0.93 and 0.80) added to Section 5.
Sun 25 Oct, Wed 28 Oct	Quyền (leader), all	Submit report and slides; presentation with live demo and backup video.
Week 45 (2–8 Nov)	Quyền, Phong	HR walkthrough with findings. Design and freeze a new benchmark of unseen, harder questions (scenario, multi-hop, conditional and temporal, paraphrased), reviewed by members.
Week 46 (9–15 Nov)	Dũng, Quyền	Run the current system and its ablations on the new benchmark; add the dense-retrieval ablation; failure analysis.
Week 47 (16–22 Nov)	Phong, Dũng	Entry-point attack tests; permission and fault checks.
Week 48 (23–29 Nov)	All	Reproduction from the recorded runs, demo hardening, final report.
Week 49 (30 Nov–2 Dec)	All	Tagged release and Round 1 presentation on 2 December.

Table 6: Weekly plan to the first final presentation, with assigned owners.

Riskiest assumption and fallback. The riskiest assumption is that a small 2–4B local controller

maintains high strict accuracy on harder, unanchored questions (paraphrased scenarios that do not name the decision or decree number) without exceeding the 2-repair budget. If strict success drops sharply on the new benchmark or dense retrieval and NLI verification do not improve over the current pipeline, we keep the bounded loop, the L1 citation verifier, the two corpora and the evaluation harness as the core deliverable, report the measured numbers transparently, and drop the optional NLI check and extra domains.

Self-assessment: on track for D3. At D2, the end-to-end system, both corpora, the 100-question evaluation suite, the 60-item human audit of the judge, and the one-command reproduction pipeline are complete and verified to run and measure reliably (57–61% strict success with the full 4B system). Significant room for system improvement and deeper experimental evaluation remains for **D3**, including the harder unseen multi-hop benchmark (W45–W46), verifier false-reject reduction, and systematic red-team attack evaluation (W47).

8 Member Contributions and AI-Use Statement

Member	Contrib.	Role in the proposal	Completed work and repository evidence
Lê Trí Quyền (Group Leader)	33%	Retrieval and serving: ingestion, BM25 + dense index, RRF, reranking, local deployment	Corpus ingestion and normalisation (`src/ingestion/`), BM25 + TF-IDF and vector indexing with RRF fusion (`src/indexing/`), cross-encoder reranker integration (`src/retrieval/`), `llama.cpp`/Prism local serving and cross-platform Docker deployment (commits `555919a`, `8fed7c5`, `15fd823`, `ad0881b`, `bdb5d69`, `415efda`).
Võ Phạm Tuấn Dũng (Technical Lead)	33%	Agent, architecture and evaluation: system design, model selection, control loop, verifier and evaluation	Idea proposal and verification, system architecture and local model selection, codebase organisation and refactoring; bounded ReAct control loop and config-driven pipeline (`src/agent/`, `src/pipeline/`), L1 citation verifier and bounded repair (`src/verifier/`), 100-task benchmarks (`evaluation/benchmarks/`), evaluation harness, GPU/Mac runs, 60-item judge audit, statistical analysis and D2 report/slides (all remaining commits in the repository).
Nguyễn Hoàng Phong	33%	System and security: tool layer, UI, audit log, persistent store, fault and injection harness	Tool registry and code-enforced permissions (`src/tools/`), regex guardrails and PII filtering (`src/security/`), Web and Streamlit interfaces with trusted outbox (`frontend/`, `src/api/`, `src/ui/`), SQLite store and audit ledger (`src/memory/`), and fault/security test suites (commits `6dc1dbc`, `78bb069`, `2c1f3d0`, `d117349`, `7ce7e45`, `1272dc7`, `e0b4a05`, `edc9bfa`).

Table 7: Member contributions (equal 33% split) matched to proposal responsibilities, repository modules and commit history.

AI-use statement. The group used a range of AI models from Anthropic (Claude), Google (Gemini, Gemma) and OpenAI (Codex) for discussion and execution support. The problem framing, system scope, design decisions, task assignment to coding agents, supervision and code review were done by the group; every change was reviewed and approved by a group member before it was committed. AI coding agents implemented code from specifications written or approved by the group. AI models were also used, under the group's direction, to draft and cross-check benchmark questions (all questions were then reviewed by a group member), to act as the evaluation judge (an LLM estimate, checked against the group's human audit), and to draft analyses and report text that the group edited.

Verified by the group. Võ Phạm Tuấn Dũng reviewed all 100 benchmark questions against the source PDFs (23 October), audited the 60-item blind judge sample (25 October: 96.7% agreement, $\kappa = 0.93$ on HCMUT; 90.0%, $\kappa = 0.80$ on labour law), verified the architecture and code across all commits, and checked the Mac demo runs. Lê Trí Quyền verified corpus extraction, BM25/TF-IDF indexing and local `llama.cpp`/reranker serving. Nguyễn Hoàng Phong verified tool permission boundaries, guardrails and Web UI states. Automated tests use real components and no mocks (425 passed, 2 skipped, 1 failed needing an unstarted Ollama backend).

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